

Incompleteness Before Arithmetic: Total Foundational Incompleteness of Every Formal System Without Self-Reference, Subsuming Gödel

The Denotation Gap, the Two-Flank Witness, and the Subsumption of the Gödelian Window in Jurisdiction, Classification, and Explanation but Never in Mechanism, via Trisduction

A scope claim that carries its own bound. The containment is proved by two classical witnesses the author does not control, and the same witnesses forbid the stronger claim the containment invites.

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Two distinct conditions are called incompleteness and the distinction is rarely drawn. One is a theorem about an interval of theory-space: for a consistent, effectively axiomatized system interpreting a weak arithmetic, there is a sentence the system neither proves nor refutes. The other is a condition of the whole line: a formal system is a discipline of strings, its derivation operator carries strings to strings, and no string is a helium atom. This paper proves the second condition, names it foundational incompleteness, and shows it needs no arithmetic, no coding, no diagonal, and no self-reference. One analytic argument establishes it, from the type of the derivation operator, and the keystone is that argument in contrapositive form: a syntax whose derivational closure contained a non-string would not be a syntax, so foundational completeness is a contradiction in terms rather than an unproved sentence. A second and premise-disjoint argument convicts grounding rather than closure, since every derivation is an actuated event and therefore an instance of the actuality it would deliver, never its source. A physical leg does no establishing work and defeats the implemented-system objection, on a content-blindness of inscription cost that is analytic once the encoding frame is fixed. A two-flank witness then fixes the scope by applying these arguments outside the Gödelian window on both sides at once, Presburger arithmetic complete and decidable beneath it and true arithmetic complete and non-axiomatizable above it, both incomplete toward the deed. The Gödelian window is thereby subsumed, and the subsumption is typed at exactly three registers and barred at a fourth. It holds in jurisdiction, since the window is a proper subclass of formal systems; in classification, since a Gödelian void is a void of strings interior to a medium whose exterior boundary the analytic argument fixes; and in explanation, since a medium that cannot deliver being is a medium whose whole operation is representation, which is the condition under which an interior diagonal is possible at all. It fails in derivation, and the paper's own Presburger witness is what forbids it: foundational incompleteness cannot entail Gödelian incompleteness, since Presburger arithmetic has the first and demonstrably lacks the second. The first incompleteness theorem is left exactly where a century of proof and four machine verifications left it. What moves is its rank, not its content. Every numerical claim in this paper reproduces by executing the battery printed in Appendix B, which requires only numpy.

KEYWORDS incompleteness, foundations, denotation, use-mention, Presburger arithmetic, true arithmetic, Landauer bound, subsumption, self-reference, reproducible verification

The conflation

The word incompleteness carries two loads in the literature and almost never carries a label saying which. The first load is precise and famous. For any consistent, effectively axiomatized theory interpreting Robinson arithmetic, there is a sentence neither provable nor refutable in it, and no such theory proves its own consistency.^{1,2} These are correct, they disclose every hypothesis, and the first has been verified in four independent proof assistants.³ Nothing here contests any line of them, and a reader who suspects otherwise should hold that fixed through every section that follows, because the argument in this paper is not with the theorem.

The second load is a condition rather than a theorem, and it is the one this paper is about. A formal system is a discipline of strings. Its alphabet is a finite set of marks, its axioms are strings, its rules take strings to strings, and its derivational closure is a set of strings. Prove that hydrogen fuses and you hold a proof. You do not hold seventeen megaelectronvolts. Prove that a bridge stands and you hold a proof. You do not hold a bridge. This gap between what a syntax delivers and what a syntax is about needs no arithmetic, no coding of syntax into numbers, no diagonal, and no self-reference. It is a fact about the medium.

Call the first condition Gödelian incompleteness and the second foundational incompleteness. The central observation of this paper is arithmetical in the simplest sense: Gödelian incompleteness is a theorem about an interval of theory-space, and foundational incompleteness is a condition of the whole line. The interval is the window of hypotheses the theorems require, consistency, effective axiomatization, and arithmetic strength. Outside that window the theorems have nothing to say and do not pretend to. Foundational incompleteness is outside no window, because its only hypothesis is that the system is made of strings, which is what it is to be a formal system.

The habit this paper isolates reads the interval's theorem as the line's verdict. It phrases the conclusion as truth outruns proof, mathematics cannot capture its own truth, no complete foundation is possible, and it deploys the phrase but Gödel proved as a terminal move in foundational argument. Each of those goes past what is established, and the distance is not rhetorical excess. It is a register error with a locatable seam, and the seam is the subject of the next several paragraphs.

Why the wider fact is invisible

If foundational incompleteness is total and Gödelian incompleteness is an interval, the obvious question is why the field's foundational reflex names the interval. The answer is not that anyone denies the wider fact. Ask any logician whether a proof of fusion is fusion and the answer is immediate and slightly impatient. The answer is that the wider fact is constitutive, and constitutive facts are invisible in the way water is invisible to a fish.

A surprising fact gets a name, a theorem number, and a literature. A constitutive fact gets absorbed into the definition of the object and then never mentioned again, because mentioning it feels like mentioning that triangles have three sides. The medium's inability to deliver being is exactly of this type. It is built into the definition of a formal system so completely that it never surfaces as a claim, and having never surfaced as a claim it never acquired a name, and having never acquired a name it never entered the ledger of things that bound formal method.

The consequence is precise. When the field reaches for the deepest limit on what mathematics can rest on, it reaches for the deepest *named* limit, and the deepest named limit is Gödel's. The invisibility of the constitutive fact is therefore not a separate problem alongside the inflation. It is the inflation's mechanism. A limit that has no name cannot be compared to a limit that has one, so the named limit inherits by default a rank that was never argued for and that its own quantifiers forbid.

This diagnosis has a testable consequence, and Section 4 discharges it against the reading that carries the reflex: if the habit is driven by naming rather than by evidence, then the argument for Gödelian maximalism should never turn on any hypothesis of the theorems, and should instead turn on the word incompleteness doing double duty. The test is narrow by design. It is a claim about the maximalist reading alone, not about every reading in the literature, and Section 4 finds that one of the readings surveyed there does turn on a hypothesis of the theorems, which is exactly why it is an argument rather than the reflex.

The barrier is structural, not computational

A barrier that yields to better computation and a barrier that requires different geometry look identical from a distance, and the distinction decides whether a research program is worth funding. This one is structural, and the demonstration is short.

Suppose someone produces a proof of the first incompleteness theorem that is sharper by any measure one likes: shorter, more elementary, formalized in a weaker metatheory, or generalized to a broader class of systems. Not one of those touches the reading habit, because the reading habit is not a claim about the theorem's proof. It is a claim about the theorem's rank. Sharpening a theorem does not change its rank, and no amount of proof-theoretic improvement inside the window says anything about the window's position on the line.

Now suppose the opposite, that someone finds an error in the proof. The reading habit would collapse, but so would the theorem, and the field would be left with the same unnamed constitutive fact it had before, still unnamed. The habit is therefore insensitive to the theorem's truth in both directions, which is the signature of a barrier that lives somewhere other than where everyone is digging.

Where it lives is the identification of a system's inability to establish its own grounding from inside with the absence of a ground to be established. That identification is a substitution of a fact about an instrument for a fact about the world, and it is licensed by an

instrument-property this paper makes formal: a magnitude blind to sign cannot orient toward anything. Section 6 proves it and Section 7 exhibits it.

The domain of validity, mapped exactly

Precision here is worth more than emphasis, so the boundary is drawn without slack. Where the question is what a fixed, consistent, effectively axiomatized system interpreting Q proves about itself, the incompleteness theorems are exact, final, and unimprovable in the relevant sense. There is no gap in that domain, no ambiguity at its edges, and nothing in this paper narrows it by a single system.

Beyond that domain the theorems fall silent, and the silence is not a verdict. Two questions in particular lie outside it. What grounds the arena in which such a system is stated, the alphabet, the coding, the provability predicate, and the metatheory in which the exhibited sentence is called true? And what does the medium of syntax do to the relation between derivation and the world it denotes, for systems inside the window and outside it alike? The first is closed in Section 6 by an argument from the theorem's own quantifiers. The second is the paper's subject.

The failure mode of the maximalist reading is a use-mention slide, and stating it in one sentence is the fastest route to the paper's core: it lets a theorem stated inside an arena speak about the existence of the arena. The slide is invisible in ordinary prose because the same words, incomplete, cannot prove, and truth, are well typed on both sides of it. That is the last thing a blind spot needs in order to survive: not obscurity, but a vocabulary that reads correctly in both registers.

What the two failure modes share

Two readings look opposite and share a root, and naming the root now saves an argument later. The inflationary reading takes the Gödel sentence as a truth the system tragically cannot reach, and concludes that formal method is permanently behind truth. The deflationary reading denies any truth above the system, takes the sentence as bare independence, and concludes there was never anything to reach. The first grants a ground and makes it unreachable. The second denies the ground and makes the drama disappear.

Both treat a reflection's inability to fix its own ground from inside as a fact about the ground rather than a fact about the reflection. The dispute between them is therefore downstream of a shared premise neither examines, and no amount of argument between them can reach it. Section 6 attacks the premise directly by exhibiting two reflections that differ precisely in whether they carry a ground, and by showing the difference is a basis-independent invariant rather than an artifact of presentation.

Prevailing readings

Six readings are surveyed. Five dominate the field and the sixth is the direct denial of this paper's first argument. Each is given its due and each is then dismantled on technical grounds. None is dismantled by preference.

The strict-formalist reading.

identifies truth with provability-in-some-system and takes the incompleteness sentences as bare independence results. It is internally consistent and it is the deflationary floor of the subject. Its technical cost is that it must then explain the informal but universal practice of calling the Gödel sentence true, a practice that appeals to the standard model of arithmetic, an object the reading declines to grant. The reading survives by paying this price, and in paying it concedes this paper's point in a different vocabulary: without a determinate ground above the ladder there is no true-but-unprovable, only independent, and the drama reduces to a statement about relative deductive strength. Note what the reading does not do. It does not deny that its own sentences are strings, and it does not claim that deriving a sentence about a bridge produces a bridge. It concedes foundational incompleteness without noticing, because the concession costs it nothing it wanted.

The naive-Platonist reading.

grants a determinate realm of mathematical truth and takes the Gödel sentence as a truth of that realm the system fails to reach. Its technical cost is the access problem: a causally inert separate realm standing in no relation to any system cannot explain how the system, or anyone, reaches it. A transcendent-only ground grounds nothing, because grounding is a relation and a relatum related to nothing is inert. The reading is not wrong about the ground's existence. It is defective about the ground's contact, and Section 6 supplies the contact the reading needs and cannot provide, by locating the ground as the fixed locus present inside every reflection rather than as a realm beside them.

The mind-exceeds-machine reading.

deploys the first theorem to argue that human insight surpasses any formal system.⁴ Its technical cost is a smuggled soundness premise: the argument requires the arguer to know the relevant system is consistent, which by the second theorem the system cannot establish about itself, and which for a system as strong as the arguer's own reasoning cannot be established without circularity.⁵ The deployment fails, and it fails where every over-reading fails, at an unnoticed re-import of a ground the argument officially lacks. Note that this reading does turn on a hypothesis of the theorems, consistency, which is precisely why it is an argument rather than a reflex, and why the test set in Section 3 is not aimed at it.

The maximalist-foundational reading.

treats incompleteness as the final word on what any formal system can rest on. This is the standing reflex that ends foundational inquiry with the words but Gödel proved. Its technical cost is exhibited in Section 6: the theorem's statement presupposes an arena, and a result presupposing an arena cannot ground it. The reading asks the theorem to do what its own quantifiers forbid. Here the test set in Section 3 discharges. The reflex never turns on consistency, on effective axiomatizability, or on interpretability of Q . It turns on the word incompleteness, and specifically on the word doing double duty across the two loads Section 3 separated. That is the signature of a

reading driven by a name rather than by a hypothesis, and it is what distinguishes the reflex from the mind-exceeds-machine argument above, which is wrong for a reason the theorems can state.

The map-territory tradition.

holds that a representation is not the thing represented, in versions running from the observation that existence is not a predicate,⁶ through the general semantics slogan that the map is not the territory,⁷ to arguments that syntactic manipulation does not constitute understanding.⁸ This tradition is correct and this paper is its descendant, which is why it is engaged here rather than cited in passing. Its technical cost is that it stops at the observation. It has no witness that steps outside the Gödelian window, so it cannot compare its own scope to Gödel's and never claims a rank. And it has no typed subsumption, so it cannot say at which registers it contains the arithmetic result and at which it does not; in the absence of that typing its adherents have historically overclaimed in one direction and its critics have dismissed it in the other. The tradition supplies the observation. What it lacks is the two flanks and the typing, and those two are what convert an observation into a placement.

The mathematical-universe reading.

is the only reading in this section that denies the paper's first argument outright, and it is therefore the one that must be met rather than surveyed. It holds that the territory is itself a mathematical or computational structure, so that the gap between a formal system and what it denotes closes at the limit.^{9,10} If the universe is a formal structure, then a sufficiently complete formal system is not describing being, it is being, and the type argument of Section 6 evaporates.

Three things are wrong with it, and the first is decisive on its own. The claim that the universe is isomorphic to a formal structure is itself a claim stated in a formal system about a referent. It is therefore an instance of exactly the relation whose existence it denies, and it inherits the gap it purports to close. This is not a rhetorical reversal but a typing fact: the isomorphism claim is a sentence, it has a truth-value, it is not the universe, and asserting it does not instantiate it. Second, the claim is unproven and is at present a posit rather than a result. It is a model-claim of exactly the kind that requires evidence, and no derivation of it exists from anything more basic. Its status is premise-grade and its adherents generally say so. Third, the reading conflates description-by with identity-to. That physics is well described by mathematics is a fact of extraordinary standing and it is not evidence for identity, since successful description is precisely the relation between a map and a territory that are distinct. The reading is not refuted here. It is located as a posit that cannot bear the weight the objection needs it to bear, which is the honest verdict rather than the convenient one.

Every reading in this section shares one structural error, and it is the error Section 6 resolves: each treats a reflection's inability to fix its own ground from inside as a fact about the ground rather than a fact about the reflection. The strict formalist reads the inability as evidence there is no ground. The Platonist reads it as evidence the ground is out of reach. The mind-exceeds-machine reading reads it as evidence that

something else reaches the ground. The maximalist reads it as evidence that no foundation is available. The map-territory tradition sees the gap and declines to type it. The mathematical-universe reading closes the gap by fiat and thereby steps into it. The instrument of Section 5 refuses the shared premise directly, by making the presence or absence of a ground a computable invariant rather than a matter of interpretation.

Method

The method is a three-condition test, stated so it reproduces without any framework vocabulary. **Triaxial verification** is the requirement that a claim about a residence, the content a proposition carries, is admitted only when three conditions hold jointly: a formal-structural reading of the residence is available, a second independent reading is available, and a third independent reading is available, the three read under a fixed transformation that leaves genuine structure invariant. Three independent readings enclosing nonzero volume is the operational meaning of a dimensionally full residence. Fewer than three, or three that collapse into a plane, is a residence the instrument reports as not full. Its operational function in what follows is to force every load-bearing claim through three disjoint routes before it is admitted, so that a claim resting on one route cannot pass by being restated on that route in different words.

The instrument is a determinant. Arrange the three readings as unit rows over a common set of contexts and form their correlation matrix. Its determinant is zero exactly when the readings are linearly dependent and positive exactly when they are independent, bounded in the unit interval. The determinant equals λ^2 , where λ is minus the signed scalar triple product of the rows, an identity binding the geometric reading, a volume, to the algebraic one, a product. Appendix A derives the sign convention and it is used throughout without variation. The instrument carries a conditioning gate so a near-degenerate residence is reported open rather than forced, and a collapse floor so a genuinely dependent residence reports broken.

Those are the three outcomes, and the **three-state verdict economy** is the discipline of using exactly them: sealed, broken, and open, with no fourth state and no fractional truth-value. A claim is sealed when its three readings are independent and a determinacy witness is supplied. It is broken when a named mechanism defeats it. It is open when the instrument finds no purchase, which is a statement about the instrument's reach and never a claim that the object is empty. The economy's operational function is to make refusal a first-class output, so that the absence of a verdict is reported rather than papered over with a hedge.

Two disciplines govern the instrument and both are load-bearing. First, a lock is permission and never proof. Three independent readings certify that a residence is dimensionally genuine. They do not certify that the proposition is true. Truth rides a separately supplied determinacy witness, and the lock only licenses its extraction. This is why no claim in this paper is sealed by the instrument alone, and why the sealed claims below each name their witness. Second, **warrant typing** is the requirement that every claim travels with its grade. Four

grades are used and no others: theorem-grade when it rests on a proof, conditional when it rests on a proof whose hypotheses are themselves posited, structural when it rests on a placement, and premise-grade when it rests on a posit. A premise stated as a theorem is a typing violation and is refused, and so is a grade the paper has not declared. Its operational function here is to keep the placement from inflating into a claim the mathematics does not carry, and it is applied to this paper's own conclusions in Section 8 with the same severity it is applied to the maximalist reading in Section 4.

Why three readings and not two or four is forced rather than stipulated. A verification algebra is finite-dimensional over the reals by construction, since it carries a finite count of reading axes. One that composes associatively, admits no zero divisors, and carries a ground identity is therefore an associative division algebra over the reals, and by Frobenius's theorem the finite-dimensional ones are exactly three: the reals, the complex numbers, and the quaternions.¹¹ The requirement of plural mutually independent imaginary axes selects the quaternions from among the three, and their imaginary part is exactly three-dimensional. Frobenius supplies the trichotomy; the axis requirement does the selecting, and neither step does the other's work. Two readings cannot close such an algebra. A fourth independent reading would demand a five-dimensional composition carrier, which does not exist over the reals as a division algebra; by the classification of normed division algebras the admissible dimensions are one, two, four, and eight,^{12,13} and at eight associativity fails. The instrument's three axes are the imaginary units of the unique associative real division algebra with plural imaginary axes, and the terminus is a closure rather than a limitation.

The Independence Verifiability Criterion is stated as a deliberate guard against verification monoculture. Every claim below is testable by independent parties using orthogonal readings. Every numerical claim reproduces by executing the battery printed in Appendix B, in full, at a fixed seed, requiring only numpy. No result rests on a datum a single authority controls, on a proprietary dataset, on a single laboratory, or on an instrument that cannot be rebuilt from the text. Where a claim rests on a classical result rather than a computation, the result is one the author does not control and did not produce, and this is not incidental: the two witnesses that fix this paper's scope and the one that bars its fourth register are all classical, all independently formalized, and all older than the framework this paper is written under.

The proposed solution

The reading inverts the order. Instead of positing a system and asking what it proves, it starts from the medium and asks what a system of that medium can deliver at all. The answer bounds every formal system whatever, and it bounds them before any question of arithmetic strength arises. The Gödelian window then appears inside that bound as an interior phenomenon, and its containment is typed at three registers and refused at a fourth.

The argument runs in two movements and the names are used throughout. **Movement I** stands on the map's shore and proves the

crossing impossible from that side for every map. **Movement II** stands on the ground's shore and shows there was never a deficit on that side. Section 6.9 composes them.

The statement

Let **RA** name the actualized substrate, the domain of things that exist, whose default condition is actuation: to exist is to carry substrate-level energetic content and to charge a cost for any transition. Let Δ_real name the operator of actualization, the assignment of physical existence to a state, which in RA is not an operation performed but the standing condition of the substrate. Let F be any formal syntactic system whatever, with its language, its axioms, and its derivation operator $\vdash_F$, under no restriction to recursive axiomatizability, arithmetic strength, or even consistency. Let $\Phi: F \rightarrow RA$ name the denotation map by which strings of F describe the substrate.

Box 1 | The theorem of foundational incompleteness

For every formal system F whatever, F is foundationally incomplete with respect to **RA**: the actuality of the substrate is not derivable, containable, or generable within F . In symbols, $\text{Perform}(\Delta_real) \notin \text{range}(\vdash_F)$. This holds independently of recursive axiomatizability, arithmetic strength, diagonalization, and self-reference. Foundational completeness of a syntactic system is not merely unachieved. It is contradictory in terms.

Warrant: theorem-grade on the type of the derivation operator and on the keystone, which is that argument in contrapositive. Theorem-grade on the content-blindness of inscription cost at fixed encoding, which does no establishing work here. Conditional where the second argument enters, a valid derivation whose hypothesis, the actuation reading of existence, is posited; it is not needed for the box.

One analytic argument establishes it. The keystone is that argument turned around. A second and premise-disjoint argument convicts a different thing. A physical leg defeats an objection. A two-flank witness fixes the scope by applying the arguments outside the window. The five do different jobs and the paper does not count them as five independent supports, because they are not.

The analytic argument: the type of the derivation operator

Every element of F has two faces and everything below turns on keeping them apart. The **token** is a physical inscription: ink, charge, or spin. The **content** is the propositional role the token carries.

The derivation operator acts on tokens and returns tokens. The closure of F under $\vdash_F$ is a set of strings, because that is what a syntactic system is: an alphabet of marks, axioms that are strings, rules that carry strings to strings. The actuality of the substrate is not a string. Therefore $\text{Perform}(\Delta_real) \notin \text{range}(\vdash_F)$. The argument is settled by the definition of a syntactic system before any physics is consulted, and it is the whole of what establishes Box 1. Theorem-grade.

The proof of fusion is a proof. It is not seventeen megaelectronvolts. Everything else in this section either turns this argument around, defends it against an objection, convicts a second thing, or carries it outside the Gödelian window.

The keystone: the same argument, contrapositive

The strongest form is not that F fails to prove a hard sentence. It is that the demand breaks category, and this is the analytic argument read backward.

Suppose for contradiction that a formal system F executed Δ_{real} , its derivational closure containing not a string denoting the substrate but the substrate. Then the closure of F under $\vdash_F$ contains a non-string, and F is no longer a syntactic system, contradicting its definition. A foundationally complete formal system is therefore a contradiction in terms, and the incompleteness is analytic at the category register.

Gödel's completeness and incompleteness theorems quantify over sentences. Foundational completeness would quantify over beings. The quantifier change is a register change. The boundary is constitutive rather than defective: logic cannot swallow reality without ceasing to be logic. The incompleteness of the map toward the territory is not a glitch, not a deficiency awaiting a stronger axiom, and not Gödel's shadow. It is the topological boundary of the symbolic medium as such, the necessary edge that makes a map a map. Theorem-grade, and the same theorem as the previous subsection.

The physics: what defeats the implemented-system objection

The type argument invites one serious objection and the physics is here to defeat it, not to establish anything. The objection: a formal system could be physically implemented such that deriving x causes x. A control system derives "valve open" and the valve opens. Does the derivation not then confer actuation on its denotation?

It does not, and content-blindness is why. The token is fully substrate-bound. It exists by actuation, it costs at least $k_{\text{BT}} \ln 2$ per irreversibly erased bit,¹⁴ and this bound is measured rather than postulated.¹⁵ Nothing in this paper calls a token weightless. The temptation is real and it is refused, because a weightless sign would contradict the very physics the argument stands on: the leg needs the inscription to cost, and to cost blindly.

Blindly is the operative word. Fix the encoding frame at a width of n bits. Landauer's bound then charges $n k_{\text{BT}} \ln 2$ for erasing the register irrespective of what the register says, so E_{token} is a constant functional of the sentence at that frame. In particular $E_{\text{token}}(x) = E_{\text{token}}(\neg x)$, and a constant carries zero information about any property whatever, truth included. In symbols, the claim at a fixed frame is $I(E_{\text{token}}(x); \Delta_{\text{real}}(\Phi(x))) = 0$. This is analytic once the frame is fixed, and Appendix B reports it as an identity rather than as a measurement.

The objection now collapses. The valve-coupling is a contingent arrangement of tokens, wired by someone, and the cost of the derivation is identical whether the valve is wired or not. The valve

opens because of the wire, not because the string denotes. Strip the wire and the derivation is unchanged, in content and in cost alike.

The qualifier at fixed encoding is load-bearing and is not protective. Appendix B reports both cases. At a fixed frame the channel is silent by construction. At a natural variable-length frame it is not silent: the recorded run returns 0.003816 bits, 0.3821 percent of the truth entropy. Any variable-length frame breaks the constancy outright and by construction, since the canonical surface string of $\neg P$ is one symbol longer than that of P, so the sign of the leak is forced by the encoder and only its magnitude is a fact about the sample. What leaks is the draftsman's convention. Section 7 states what that does and does not establish.

The structural rhyme with the instrument is stated at the level where it is sound. No cost fibre determines a truth-value, since every fibre is stable under negation at a fixed frame; and the lock scalar of Section 6.7 determines no truth-sign, since it is invariant under the reflection that flips the sign. Both are functionals that cannot carry what their own symmetry destroys. The instrument's half of the rhyme is exact and pointwise. The cost half holds at the possibilistic level just stated and is not an information-theoretic parallel, for the reason Section 7 records.

The second argument: grounding, not closure

This one is premise-disjoint from the type argument and it convicts a different thing. The type argument convicts closure: the substrate is not in $\text{range}(\vdash_F)$. This argument convicts grounding: no derivation can be the source of the actuality it asserts.

Every token of F is itself an actuated existent inside RA. The map is a proper physical part of the territory it describes, and every act of derivation is an event the territory hosts. So consider any derivation D inside F purporting to establish $\Delta_{\text{real}}(\text{RA})$, that the substrate is actual. D is itself an actuated event. Its tokens exist, its steps cost, its inscription rides the substrate. The derivation therefore presupposes, in its very performance, the actuality it purports to deliver. The presupposition is material rather than diagonal: no coding, no fixed point, no self-referential sentence, only the fact that proving happens in the world. A proof of the world's actuality is an instance of the world's actuality and hence never its source. The attempt expends the very substrate it would generate, so the attempt certifies the substrate and cannot ground it.

The scope must be stated before a hostile reader states it: this argument convicts delivery, not reference. A string can refer to actuality perfectly well; what it cannot do is instantiate it. This paper refers throughout and instantiates nothing, and Movement I applies to it without paradox, since Movement I does not deliver the substrate's actuality but observes that no derivation delivers it, and that observation is a negative result about reach stated at the level of reach. Conditional: the derivation is valid and its hypothesis, the actuation reading of existence, is posited, so the conclusion holds at exactly that premise's strength. It is not needed for Box 1, which the type argument already carries.

The two flanks: carrying the arguments outside the window

Gödel's theorems govern exactly the window $H(F)$: F consistent, effectively axiomatized, interpreting Robinson arithmetic Q . This subsection does not establish foundational incompleteness. It applies the arguments above to two witnesses whose position relative to $H(F)$ is fixed by two classical completeness results, and it thereby fixes the scope. Taking one witness on each flank is the whole point, since a single witness on one side would leave the other open to the reply that the window's edge is where the wider fact begins.

Beneath the window. Presburger arithmetic, the first-order theory of the natural numbers with addition and without multiplication, is consistent, complete, and decidable.^{16,17} It cannot interpret Q , so it fails the antecedent of the first incompleteness theorem. Gödel's theorem, being a universally quantified conditional, is therefore vacuously true of it; its conclusion is what is false of Presburger, since Presburger is complete. That is the sharper of the two facts and it is the one Box 6 uses. Presburger decides every sentence of its language. It is syntactically complete in the fullest sense available to it. And it actualizes nothing whatever. It is foundationally incomplete by the type argument.

Above the window. True arithmetic $Th(\mathbb{N})$, the set of all sentences true in the standard model, is consistent and complete by construction and is not recursively axiomatizable.¹⁸ It fails the effective-axiomatization hypothesis, so the theorems do not reach it either. It contains every true arithmetic sentence. It is the maximal syntactic completeness conceivable in its language, a theory that has, in the only sense the phrase can bear, everything. And it remains ink, bits, and denotation. It is foundationally incomplete by the same type argument.

Even an imagined hyper-computational system, an oracle theory in which every true syntactic statement is proved, fails identically, and it fails for a reason worth stating in its own words: its completeness is completeness of the truth-set, and foundational completeness would be completeness of being. The second does not follow from any amount of the first, and no strengthening along the first axis moves a millimetre along the second, because the two axes are not the same axis.

Gödelian incompleteness lives on an interval of theory-space. Foundational incompleteness is total over the whole line. That is the exact content of the word non-Gödelian in this paper's thesis, and it is earned as a classification rather than worn as a slogan.

The dependency ledger

Gödel-Rosser requires $H(F)$ and constructs a self-referential sentence G_F . Movement I requires neither H nor any fixed point, and the dependencies are stated exactly.

The type argument rides the definition of a syntactic system and nothing else. The keystone rides the same definition and is the same argument. The physics leg rides Landauer's bound and a fixed encoding frame, and it defends rather than establishes. The material

self-presupposition argument rides the actuation reading of existence and the nesting of the map inside the territory; it is premise-disjoint from the type argument and it convicts grounding rather than closure. The two-flank witness applies the type argument to two systems whose position outside $H(F)$ is fixed by Presburger's completeness theorem and the non-axiomatizability of $Th(\mathbb{N})$; it is not premise-disjoint from what it applies, and it is not offered as independent support.

So the honest count of premise-disjoint arguments for Box 1 is two, not three, and only the first of the two is needed. No leg invokes arithmetization, provability predicates, or diagonalization, and the result holds where Gödel's theorem is vacuously true of the system and its conclusion false, Presburger, and where the theorem is inapplicable, $Th(\mathbb{N})$. The incompleteness is an artifact of the medium, not of arithmetic loops.

The instrument: locating the diagonal inside the medium

The arguments above bound every formal system. They do not by themselves say what the Gödelian phenomena are, where they sit, or why a medium of the kind described should host them at all. For that the instrument enters, and it enters subordinated: it is not the thesis, it is the apparatus that locates the thesis's interior structure.

Fix a reflection, an operation squaring to the identity. On the quaternions the natural reflection is conjugation, which fixes the real line and negates the three imaginary axes. Its fixed locus is one-dimensional, the ground, and the space it negates is the three-dimensional residence. This is the ground-bearing reflection. Collapse the real line and the same operation degenerates into the antipodal involution, negation on the whole space, whose fixed locus is the origin alone and therefore has dimension zero. This is the engine of self-referential incompleteness: the diagonal construction that produces an object disagreeing with every row of a list along the diagonal, which in categorical form unifies Cantor's, Russell's, Tarski's, and Gödel's results as one fixed-point phenomenon.¹⁹

Everything below rests on the observation that these two reflections are genuinely different objects, and the four results make it formal.

Box 2 | Theorem 1 (Fixed-Locus Invariance)

Let T be a linear involution on a real vector space V , so $T^2 = I$. Then V is the direct sum of the plus-one eigenspace, the fixed locus of T , and the minus-one eigenspace, and the dimension of the fixed locus is a similarity invariant: for any invertible h , $\text{Fix}(hTh^{-1}) = h(\text{Fix } T)$, hence of equal dimension.

Proof. Since $T^2 = I$, the minimal polynomial divides $t^2 - 1$ with distinct roots, so T is diagonalizable with eigenvalues in $\{+1, -1\}$, giving the direct sum. If $Tv = v$ then $hTh^{-1}(hv) = hv$, so h carries $\text{Fix } T$ isomorphically onto $\text{Fix}(hTh^{-1})$, preserving dimension. For conjugation on the quaternions the fixed locus is the real line, dimension one. For the antipodal involution, negation on the same four-dimensional space, the fixed locus is the origin alone, dimension zero. The two involutions are therefore not conjugate and not interchangeable by any change of basis. ■

The splitting is stated as a direct sum and not an orthogonal one, since orthogonality would require T self-adjoint and the self-adjoint involutions are not closed under the similarity this theorem quantifies over. Nothing here needs it: dimension is the invariant, and both involutions the paper uses are self-adjoint in any case.

Warrant: theorem-grade.

Theorem 1 makes carrying a ground and carrying no ground a basis-independent structural fact rather than an artifact of presentation. The diagonal engine of self-reference is the dimension-zero involution. It carries no ground, so it founds nothing at the ground layer, and a construction powered by it bounds only the ladder it runs on.

Box 3 | Theorem 2 (Orientation-Blindness of the Lock Scalar)

Let three unit rows form a correlation matrix R and let $\det R$ be the lock scalar. Then $\det R = \lambda^2$, where λ is minus the signed scalar triple product of the rows, and $\det R$ is invariant under reflection of any single row. In particular the locking verdict on a residence equals the locking verdict on its reflection, so the scalar certifies the dimension of the residence and never its sign.

Proof. Reflecting one row multiplies it by -1 , acting on the correlation matrix as conjugation by a diagonal sign matrix D with a single -1 entry. Then $\det(DRD) = \det(D)^2 \det(R)$, and $\det(D)^2 = 1$, so the determinant is unchanged. The determinant equals λ^2 and squaring discards the sign. Hence the verdict, a function of the determinant, is identical for a residence and its reflection. ■

Warrant: theorem-grade.

Theorem 2 is why symmetry-only reasoning cannot settle a directed question, and it is the exact formal reason a ground-blind scalar cannot orient toward a ground on its own: the sign a decision would need is precisely the information the square destroys. Recovering direction requires reading the ordered axes directly, never the bare

magnitude. This is not a defect of the instrument. It is a theorem about which questions a magnitude can answer.

Box 4 | Proposition 3 (Imprint Classification)

Say a direction is **field-permitted** when the instrument returns a lock on it: its three readings are populated and independent. Note that a direction on which the instrument returns open is not thereby shown unpermitted; it is a direction on which the instrument reports no purchase, and the classification below is stated for archetypes in which the negation is stipulated contentless rather than found so.

Run the instrument on a proposition P and on its negation. Read the residence as **imprinted**, grounded, when exactly one direction is field-permitted and a determinacy witness is supplied. Read it as a **ghost**, ungrounded, when both directions are field-permitted, certified for a genuine independence result by a supplied independence proof. Then a proven theorem and a true-but-unprovable sentence return the identical grounded verdict, differing only in whether a rung, a constructed proof, exists; a genuine independence result returns the ghost verdict and is a categorically different object.

Argument. A proven theorem has determinate content its negation lacks, so exactly one direction is field-permitted with a witness, the proof, in hand. A true-but-unprovable sentence likewise has determinate content its negation lacks, established by the metatheoretic argument that shows it true, so exactly one direction is field-permitted with a witness, the metatheoretic argument, supplied, while no rung inside the system exists. Both read grounded. A genuine independence result has both directions field-permitted, certified by the independence proof, and reads as a ghost. ■

Warrant: structural. The archetypes of Appendix B exhibit the three cases as mechanically distinct; they do not establish that any particular sentence is of one type rather than another, since each archetype's negation is constructed contentless by hand.

The corollary is the over-reading the placement most sharply forbids. Reading a formal sentence as groundless noise, a ghost to be evicted, misclassifies an imprinted object as a ghost. The Gödel sentence is grounded and rung-absent, not groundless, read exactly as a proven theorem is, the difference living only in the ladder above.

Box 5 | Proposition 4 (Arena-Presupposition)

Let a theorem have a statement whose well-formedness requires an arena: an alphabet, a coding of syntax into arithmetic, a provability predicate, and a metatheory in which the relevant truths are established. Then the theorem does not entail the existence conditions of that arena, and no ground that supplies the arena is derivable from the theorem.

Argument. The theorem's quantifiers range over objects defined only relative to the arena, so its content is a predicate on arena-relative objects and asserts nothing about the existence of the arena, on pain of a use-mention conflation. A conditional whose antecedent presupposes the arena cannot have the arena's existence as a consequent without circularity. Hence a theorem that measures the reach of a provability predicate, itself an arena-relative object, cannot be the ground of the arena that gives the predicate sense. ■

Warrant: structural.

Proposition 4 is the formal core of the title's first half. It places the incompleteness theorems inside the stratum they describe. They measure the ladder and presuppose the arena the ladder is built in. They therefore neither decide the truth of the sentences they exhibit, which the metatheory decides, nor ground systemhood, which their own quantifiers forbid. They are neither dictator nor final guard.

One cited result completes the interior picture, and it is the sharpest observation available that touches the mathematics rather than its reading. For a consistent effectively axiomatized theory extending Peano arithmetic there exist two provability predicates, both numerating the axiom set extensionally, such that the theory does not prove the consistency statement built from the first, which satisfies the standard derivability conditions, yet proves a consistency statement built from the second, a deviant numeration failing them.²⁰ The truth value of the sentence *a system proves its own consistency* is therefore not invariant under the choice of provability predicate. A token that is an output of a chart-selection rather than of the underlying structure may not enter a verdict as a structural fact without the chart named, and the consistency token is such an object. The chart-independent invariant is the first theorem's reach fact. The second theorem's headline carries a chart-dependent token, and the honest statement names the chart, the derivability conditions, under which it holds. This refines the theorem's scope and contradicts nothing; the second theorem is machine-verified under its stated conditions, and the observation has been available for sixty-six years.

Two placements close the interior, and the first is stated with its scope restricted. Where independence is measured by consistency strength, the diagonal re-enters at the root through the provability predicate of the second theorem, and Paris-Harrington at ϵ_0 is the clean witness: its combinatorial statement is not self-referential, and its unprovability in Peano arithmetic runs through a fast-growing function that dominates every provably recursive function of the theory and hence yields $\text{Con}(\text{PA})$.²¹ Not all independence is of this form. Kruskal's theorem and $\text{TREE}(3)$ sit beyond Γ_0 by ordinal strength,²² and Borel

determinacy over Zermelo set theory is carried by the height of the cumulative hierarchy, requiring $\aleph_1$ iterations of the power set.^{23,24} Whether every such case reduces to a consistency statement is not settled here and no unrestricted claim is made. What is established is narrower and sufficient: the diagonal drives self-referential incompleteness, where the fixed-point theorem lives, and natural non-self-referential statements are independent by proof-theoretic strength rather than by self-encoding. The second placement is that the provable stratum is graded, not merely bounded: reverse mathematics resolves provable-in-T into a strength-location on a well-ordered tower of subsystems, each theorem pinned to its minimal subsystem and proof-theoretic ordinal.²⁵ The coarse provable-or-not becomes a rung-location. The ladder falls short, and reverse mathematics measures by how much at each rung.

The dissolution at the anchor

Movement I stands on the map's shore. Movement II stands on the other and finds there was never a deficit there to begin with. Four steps, each stated at its own grade.

The domain boundary is exact. The limitative theorems bind precisely on $H(F)$: consistent, effectively axiomatized, interpreting Q . Gödel-Rosser yields the undecided sentence, the second theorem the unprovable consistency statement, Tarski the undefinable truth predicate,²⁶ each in its well-typed form. Inside that window they bind with full force and this paper narrows none of it.

They do not attach outside it, and the non-attachment is a parse rather than a conquest. The substrate is not a sentence-set, not recursively enumerable, and not a candidate for a consistency predicate. It actuates and interprets nothing. The theorems are not false at the territory. They are vacuous there, and the sentence *reality is incomplete* is ill-typed, transporting a predicate defined on ladders onto a domain that is not one, with no bridge to license the transport, because the claim that the universe is isomorphic to a recursively enumerable theory is itself the unproven model-claim dismantled in Section 4. The theorem never sat on the throne of reality, because the throne is not in its domain.

The anchor's warrant is registration rather than derivation. That existence carries energetic content is theorem-grade external physics for confined quantum existents: the kinetic-energy lower bound and the zero-point energy follow from the canonical commutation relation, mass-energy equivalence fixes positive rest energy, and the rigorous third law bars a cooled-to-zero state.²⁷ Incompleteness is a property of theories. The anchor is not a theory. And a posited foundation cannot be promoted to a theorem of its own base: the underivability is constitutive of foundation-hood and is itself a result about foundations rather than a foundation, so the exemption is earned rather than claimed.

Truth-adjudication re-registers. For a string the ladder cannot reach, adjudication moves from the provability predicate to the classification of Proposition 3. The Gödel sentence is ground-present and rung-absent. The unreachable truths are unbuilt rungs and never missing ground.

The determinacy of any particular arithmetic string rides a premise, that the standard model has a determinate truth-value for it, and that premise is not established here and is not established anywhere. It is named, it is held at premise grade, and it caps every structural claim in this movement. It is not crossed.

The join: one boundary, two shores

The two movements are one boundary read from its two sides, and their composition is this paper's content rather than their conjunction.

Movement I proves the crossing impossible from the map's side for every map: foundational incompleteness total over the line, analytic at the type argument and its keystone, defended at the Landauer leg, doubled at the presupposition leg, and carried outside the Gödelian window on both flanks. Movement II proves there was never a deficit on the ground's side: the theorems vacuous off the ladder, the anchor registered, the voids reading as imprints.

Held together they resolve the tension the words alone create, that a thing can be at once truly incomplete and dissolved of incompleteness. The incompleteness is a predicate of maps and it is total there. The determinacy is a condition of the ground and it is native there. An architecture anchored on the ground side carries its formal engine as an instrument whose measured shortfall testifies for the ground it reaches toward.

One reconciliation is stated so no seam remains. Every precisifying deed is an actuation and pays its cost. The physics leg observes that the deed's charge never transfers to the denotation. These are two faces of one coin: every proof pays for itself and pays nothing toward its referent, the deed actuated and the content inert, which is exactly why the attempt to capture the ground certifies it instead.

The subsumption, typed

Three containments hold. One entailment is barred. The hierarchy is thereby exact, and the bar is what makes the containments credible rather than triumphal.

Containment one, jurisdiction. Gödel's theorems govern exactly $H(F)$, a proper subclass of formal systems. Foundational incompleteness governs all F whatever. So $\text{dom}(\text{Gödel}) \subsetneq \text{dom}(\text{Foundational})$, and the properness is exhibited rather than asserted: Presburger arithmetic beneath the window and $\text{Th}(\mathbb{N})$ above it are both inside the wider frame's jurisdiction and both outside the sub-module's reach. Every system Gödel touches is already inside the frame's jurisdiction, and the frame touches systems Gödel cannot. **Theorem-grade** on the domains.

Containment two, classification. Every Gödelian void is a void of strings: an unreachable string inside a string-discipline. Movement I establishes that strings are all the closure ever contains, and fixes the medium's exterior boundary. The Gödelian void is therefore located inside the medium whose exterior boundary Movement I fixes, not beside it. The two are not nested as instances and no such nesting is claimed: foundational incompleteness is one condition per system, indexed by F , and asserts that a non-string is absent from the closure; a

Gödelian void is indexed by a system and a sentence, and is a string absent from the closure. They are an interior void and an exterior boundary of the same medium, and that relation is what makes the location a containment rather than a coincidence. **Structural.**

That this is a register distinct from the third, and not a corollary of it, is settled by what the two claims assert rather than by any witness. Containment two asserts a location relation: the module's voids sit interior to the medium whose exterior boundary Movement I fixes. Containment three asserts an explanation relation: the medium's boundary condition is why the module's mechanism has room at all. Location and explanation are different relations, and two claims asserting different relations between the same relata are different claims whether or not any system separates their truth-values. None does, and none is offered: at Presburger arithmetic both are idle, the second because a complete theory has no voids to locate and the third because Presburger lacks the richness Box 6 names, and a vacuity that falls on both sides discriminates neither.

Containment three, explanatory precedence. The wider frame is prior in the order of explanation. A medium that cannot deliver being is a medium whose entire operation is representation. A representation-discipline rich enough to represent its own representing can knot on itself, and the diagonal is the canonical interior knot. The boundary condition therefore explains why interior limitative phenomena are possible, locates them as interior, and strips them of foundational rank. It does not make them expected, and Presburger is why: the frame covers media in which the knot never forms. The sub-module operates on the diagonal of provability predicates inside the ladder, a valid, accurate, and internal certificate of the map's self-referential bounds: the ceiling technician of the map's interior and never the architect of the building, standing untouched inside the frame that seats it. **Structural.**

Box 6 | The barred entailment

Subsumption of scope is not derivation of mechanism. The Gödelian mechanism, arithmetization, the provability predicate, the fixed point, is not a consequence of the medium-limit and cannot be derived from it. This paper's own flank decides it in one line.

Presburger arithmetic is foundationally incomplete by Movement I and syntactically complete by Presburger's classical result. So if foundational incompleteness entailed Gödelian incompleteness, Presburger arithmetic would be syntactically incomplete, contradicting a completeness theorem this paper does not control and did not produce.

The derivative-consequence reading is therefore **broken**, and it is broken by the same witness that establishes containment one. The tempting bridge, that a medium rich enough to represent its own representing can diagonalize, fails at the words *rich enough*: richness is an extra hypothesis, and Presburger arithmetic is a medium that lacks it. The frame predicts room for the diagonal. It does not generate it.

Verdict: broken, with the named mechanism. Warrant: theorem-grade, on Presburger's completeness.

The honest hierarchy is then stateable in one sentence. The frame contains the module's jurisdiction, classifies its voids, and explains its possibility. The module supplies its own mechanism, which the frame

predicts room for and does not generate. Anchored outside the ladder, the arithmetic limit becomes a fixed, observable feature of the instrument, read from outside, located by the frame, graded on the reverse-mathematics tower, and measured by the consistency regress, whose rung costs exactly the height it gains.^{28,29} The internal incompleteness of the logic is framed and located by the external incompleteness of the map relative to the real. The mechanism remains the theorem's own. That is what subsuming Gödel means here, and it is the whole of what it means.

Reproducible verifications and falsification conditions

Every quantitative claim in this paper reproduces by executing the battery printed in full in Appendix B at seed 20260622 over 24 reading-contexts in double precision, requiring only numpy. The battery is printed rather than described, because a claim of reproducibility discharged by a snippet that does not run is not a claim of reproducibility. Nothing below is imported from another document. The Independence Verifiability Criterion is met because each check is an orthogonal reading no single authority controls, and because the three claims that carry the most weight in Section 6, the two flanks and the bar, rest on classical results the author neither produced nor controls.

The anchor battery

Table 1 | The recorded anchor battery, reproduced by Appendix B. Each row is an independent check. Failure on re-execution falsifies the stated result. The last two rows are typed as identity and measurement respectively, and the distinction is the subject of Section 7.3.

check	object under test	recorded result	reading	type
Fixed locus	conjugation, $\text{diag}(1, -1, -1, -1)$	$\dim \text{Fix} = 1$; involution residual 0.000e+00	carries a ground	computed
Fixed locus	antipodal involution, $-I$ on the same space	$\dim \text{Fix} = 0$; involution residual 0.000e+00	carries no ground	computed
Orientation	reflect one axis of an independent triad	$\det 0.999627146061$ before and after, difference 0.000e+00	magnitude blind to sign	computed
Orientation	the same reflection, signed scalar	$\lambda + 0.999813555650$ to -0.999813555650 , ratio -1.000000	sign lives in the axes	computed
Identity	λ^2 against $\det R$ on the same triad	residual 1.110e-16	volume equals product	computed
Classification	a proven-theorem archetype	one direction locks, $\det 0.586297$; negation stipulated contentless	imprint, grounded	computed
Classification	a true-but-unprovable archetype	one direction locks, $\det 0.588777$; negation stipulated contentless	imprint, grounded	computed
Classification	a genuine-independence archetype	both directions lock, $\det 0.617430$ and 0.596375	ghost, ungrounded	computed
Denotation	cost against truth, fixed 64-bit frame	cost takes one value, {64}; $I(E_token; \text{truth}) = 0.000000000000$ bits	constant channel	identity
Denotation	cost against truth, natural variable frame	$I = 0.003816$ bits, 0.3821 percent of $H(\text{truth}) = 0.998664690729$	channel not silent	measurement

The orientation construction

The orientation check runs on a basis fixed once by QR and reused after the reflection, so that the coordinates before and after are expressed in one frame. The full listing is in Appendix B; the decisive lines are these.

PYTHON

```
Bfix = np.linalg.qr(Q.T)[0][:, :3].T      # basis fixed
ONCE, reused after reflection
def lam(Qm):
    co = Qm @ Bfix.T; q = [np.concatenate(([0.0], c)) for
c in co]
    return float(qmul(qmul(q[0], q[1]), q[2])[0])
Qr = np.diag([-1., 1., 1.]) @ Q          # reflect one
axis: P -> not-P on that road
det0, detr = float(np.linalg.det(Q @ Q.T)),
float(np.linalg.det(Qr @ Qr.T))
lam0, lamr = lam(Q), lam(Qr)
# det0 = 0.999627146061 ; |det0 - detr| = 0.000e+00
# lam0 = +0.999813555650 ; lamr = -0.999813555650 ; ratio
= -1.000000
# |lam0**2 - det0| = 1.110e-16
```

Both quantities are computed and both are reported, since the determinant's invariance and λ 's inversion are two separate facts and the first without the second exhibits nothing.

What the denotation battery establishes, and what it does not

The two denotation rows in Table 1 are typed differently on purpose, and the typing is the finding.

The fixed-frame row is an identity, not a measurement. At a fixed 64-bit frame the cost functional takes exactly one value, and the battery reports the value set as $\{64\}$ so the reader sees this directly. A constant random variable has zero entropy, and the mutual information between a constant and any other variable is zero for every joint law whatever. The recorded 0.000000000000 bits is therefore analytic in the encoder's choice of frame. It confirms that Landauer's bound charges by register width and not by content, which is what the physics leg of Section 6.4 needs and all it needs, and it is reported as an anchor rather than as evidence. The count of 14552 formula-model pairs is reported for completeness of the construction and carries no weight in this row: one pair would return the same zero.

The evenness route was never available on this sample, and the recorded entropy is what records it. An argument from evenness would run: the cost is even under negation, the truth-value is odd under it, therefore the mutual information vanishes. That inference is invalid without a further hypothesis. Take the sample $\{A, \neg A, B\}$ with cost even under negation and truth odd; then $H(\text{truth}) = 0.918296$ bits and $I(\text{cost}; \text{truth}) = 0.251629$ bits, strictly positive. The correct lemma requires the joint law to be negation-invariant, and that

hypothesis forces $H(\text{truth}) = 1$ exactly. The battery records $H(\text{truth}) = 0.998664690729$ bits, which is strictly less than one, and the shortfall is not noise: the sample is not closed under negation, and the printed generator is why. The generator never emits a double negation at any depth, so the 682 admitted tokens that already carry a leading negation have negations outside the grammar's range outright; of the remaining tokens lacking their negation, 123 would exceed the 64-bit admission filter by the added symbol and 399 have it inside the range and simply undrawn, for 1204 of 1819 admitted tokens lacking their negation, the three categories disjoint by that order of precedence. Of 14552 pairs, 7589 are true where closure would force 7276, an excess of 313. The battery therefore discloses its own sample's asymmetry rather than relying on a symmetry it does not have.

The variable-frame row is the only one with empirical content, and its sign is not empirical. At a natural variable-length frame the mutual information is 0.003816 bits, 0.3821 percent of the truth entropy. That the leak is nonzero is forced rather than found: the canonical surface string of $\neg P$ is one symbol longer than that of P , so $\text{cost_var}(\neg P) = \text{cost_var}(P) + 4$ for every P in the sample, and no variable-length frame of this kind can be even. Only the magnitude is a fact about the sample. The channel through which it leaks is length: the battery records a correlation of +0.1310 between formula length and the frequency with which a formula comes out true across valuations. Why length should correlate with truth-frequency at all is a fact about the drawing distribution and the operator mix, and this paper does not establish the route.

The consequence, stated plainly. An unconditional claim that inscription cost carries zero information about truth is **false**, and the variable-frame row refutes it. The claim holds at fixed encoding, where it is an identity. What leaks at a variable frame is the encoder's convention, not the denotation relation, and the demonstration of that is not the mutual information but the observation that a coder who fixes the frame closes the channel completely, which is not something one can do to a genuine information channel.

One modelling step is flagged rather than buried. Section 6.4 states the content-blindness as $I(E_{\text{token}}(x); \Delta_{\text{real}}(\Phi(x))) = 0$, a claim about the actualization of what x denotes. The battery measures $I(E_{\text{token}}; \text{truth of a propositional formula under a valuation})$. Truth in a valuation is not $\Delta_{\text{real}}(\Phi(x))$, and the substitution is a modelling step, not an identity. What transfers is the structural point that the cost functional is constant across a fibre on which the semantic value varies; what does not transfer is any quantitative claim about physical actualization, which this battery does not measure and does not purport to.

Falsification conditions

Eight conditions are stated with the same precision as the results, each of them a single decisive observation rather than an accumulation of evidence.

One. The Fixed-Locus claim fails if any change of basis makes the antipodal involution acquire a nonzero fixed locus, or strips conjugation of its one-dimensional one. Neither can occur, since

dimension is a similarity invariant, so any demonstration otherwise refutes Theorem 1 outright.

Two. The Orientation-Blindness claim fails if a single-axis reflection is ever recorded changing the determinant on a fixed external basis. The recorded change is bit-zero.

Three. The Classification claim fails if a proven theorem and a true-but-unprovable sentence ever return different grounding verdicts, or if a genuine independence result ever returns anything but the ghost verdict. The archetypes of Appendix B are executable and the condition is testable against them by any party. What the archetypes cannot decide, and what this condition therefore does not test, is whether a given sentence is of one type rather than another, since each archetype's negation is built contentless by hand.

Four. The content-blindness of cost fails at its stated scope if any fixed encoding frame is exhibited for which $E_token(x) \neq E_token(\neg x)$. The frame must be fixed; the variable-frame leak reported above is not a counterexample and is disclosed here precisely so that it cannot be presented as one later.

Five. The two-flank witness fails if Presburger arithmetic is shown to be syntactically incomplete, or if true arithmetic is shown to be recursively axiomatizable. Both are settled classical results and both are formalizable in any proof assistant that carries the relevant theories, which is what makes this the strongest pre-registration in the paper: the flanks are checkable by parties who have never heard of it.

Six. The type argument and its keystone fail if a formal system is exhibited whose derivational closure contains a non-string while the system remains a formal system. This is the condition under which the paper's central claim becomes a category error rather than a category argument.

Seven, and this one is aimed at the paper's own thesis. The subsumption fails at the derivation register, in the direction of *more* rather than *less*, if foundational incompleteness is shown to entail Gödelian incompleteness. Such a demonstration would inflate this paper's containment from three registers to four. It would also contradict Presburger's completeness theorem, so the same witness that establishes the containment forbids its strengthening. A scope claim whose own witness bounds it from above is the structure this paper is offering, and this condition is where the offer is testable.

Eight, structural rather than numerical, stated as a standing prediction. If the provable stratum is genuinely graded rather than binary, then theorems independent of a weak base should sort into a strictly ordered tower of minimal subsystems with distinct proof-theoretic ordinals, and no theorem should require a subsystem it does not use. Reverse mathematics confirms exactly this sorting across a large body of ordinary mathematics. A single theorem found to require strictly more or less than its established minimal subsystem falsifies the grading claim, and the prediction is independently checkable in any proof assistant that formalizes the subsystems.

The verifications are decentralized by construction. Each is a short self-contained computation printed in Appendix B, a citation to an

independently formalized classical result, or a check runnable in a proof assistant. None depends on a proprietary dataset, a single laboratory, or an unreproducible instrument. A reader who distrusts the printed numbers executes the battery. A reader who distrusts the two flanks checks them against Presburger 1929 and the standard non-axiomatizability of true arithmetic. A reader who distrusts the grading claim checks it against the reverse-mathematics literature. This is the deliberate answer to a verification monoculture: the results survive only if they reproduce on substrates the author does not control.

Discussion and implications

The placement reframes several adjacent questions, and it is here that the reader sees what the results add up to.

The incompleteness engine is ground-blind by construction. Its reflection has a fixed locus of dimension zero, and even the magnitude it produces cannot carry the sign a grounding decision needs. A construction powered by this engine measures the ladder and is constitutively silent about the ground it measures against. This is not a wound in the first theorem. It is a placement of the theorem inside the stratum it describes.

The maximalist reading is not merely an exaggeration. By Proposition 4 it asks a theorem to do what its own quantifiers forbid, ground the arena they presuppose. And the strict-formalist deflation and the naive-Platonist inflation are two responses to one gap: neither the ladder nor a transcendent-only realm supplies a ground in contact with the system. The ground-first reading supplies exactly that, a ground the instrument reaches by reading rather than by climbing, and the access problem does not arise because the ground is the fixed locus present in every residence rather than a separate inert realm.

The second theorem is less robust than its reputation. Its self-consistency headline is chart-dependent, and the honest statement names the chart. This is the sharpest observation in the paper that touches the mathematics itself rather than its reading, and it has been available for decades.

Foundational inquiry does not end where the reflex ends it. The practical consequence of the placement is a research program rather than a conclusion, and each line of it is concrete and checkable: classifying propositions by whether they are grounded, provable, and computed rather than by provability alone; grading the provable stratum by proof-theoretic strength rather than by the coarse provable-or-not; and separating genuine independence, the ghost, from grounded unprovability, the imprint, by a mechanical test rather than by intuition. The last of these has an immediate application. If the partition is by ground-presence rather than by cardinality, then importing a maximality axiom to assign the continuum a determinate value treats a ghost as an imprint, and the move is a category error rather than a bold conjecture.

The limitation, stated symmetrically

The same instrument that convicts the maximalist reading must convict any reading that over-reaches, including this one, and the conviction is entered here rather than left for a referee.

The ground-first reading rests on a posit: that the ground is a definite object above the ladder. That posit is premise-grade, not a theorem. A reader who declines it, the strict formalist of Section 4, reads the placement as itself resting on an unproven commitment, and that reading is not refuted here. The honest position is that the placement holds at exactly the strength of the ground-first premise and no higher, and that this premise is the one every foundation must post and cannot prove, because a foundation provable from its base would not be a foundation. This paper convicts the maximalist reading of importing a ground it denies. It claims no exemption for its own ground from the same demand.

Three further limits belong here for the same reason. The material self-presupposition argument rides an actuation reading of existence which is theorem-grade at its physics floor for confined quantum existents and premise-grade in its universal extension to all existents whatever; the extension is posited, not proved, and its premise-status is not a concession but a structural necessity, since a foundation derivable from a weaker base would not be the foundation. The argument built on it is thereby the paper's conditional-grade claim, a valid derivation whose hypothesis is posited, holding at that hypothesis's strength. Movement II's re-registration of unreachable truths as grounded rides the determinacy of the standard model, which is named, capped, and not crossed. And the classification archetypes of Appendix B exhibit that the three cases are mechanically distinct; they do not and cannot establish that any given sentence falls under one rather than another, because each archetype's negation is stipulated contentless rather than found so. Neither limit is repaired by anything in this paper, and a reader who rejects any of them reads the corresponding claim at reduced strength rather than at zero.

What is not limited is the containment. Containment one is theorem-grade and turns on two classical results. The barred entailment is theorem-grade and turns on one. The premises above govern the reading of the boundary, not the arithmetic of the flanks.

Conclusion

Two conditions are called incompleteness. One is a theorem about an interval of theory-space, exact inside its window and silent outside it. The other is a condition of the whole line, a fact about what a discipline of strings can deliver, and it needs no arithmetic, no coding, no diagonal, and no self-reference. One analytic argument establishes it, from the type of the derivation operator, and the keystone is that argument in contrapositive: a syntax whose derivational closure contained a non-string would not be a syntax. A second and premise-disjoint argument convicts grounding rather than closure. A two-flank witness carries both outside the Gödelian window on either side.

The wider condition subsumes the Gödelian window at three registers and at exactly three. In jurisdiction, since the window is a

proper subclass and two classical witnesses sit outside it on opposite sides. In classification, since a Gödelian void is a void of strings interior to the medium whose exterior boundary Movement I fixes. In explanation, since a medium whose whole operation is representation is the condition under which an interior diagonal is possible at all. It does not subsume at the register of derivation, and the paper's own Presburger witness is what forbids it. A scope claim that carries its own bound is what is on offer here, and the bound is not a courtesy: it is enforced by a completeness theorem from 1929 that the author does not control.

None of this touches the first incompleteness theorem, which stands where its machine-checked proof leaves it. What moves is its rank, not its content, and the rank was never argued for in the first place. It was inherited by default from the fact that the wider limit had no name.

The single most important open question the placement raises is whether the ground-first premise can be given any warrant beyond the premise-grade it carries here, and the honest answer is that it is exactly as secure as the choice every foundation makes and no more. The reframing that acceptance would entail is small to state and large in consequence: a shortfall of proof against truth is a measurement of the ladder read from the ground that exceeds it, and not a ceiling over what mathematics may stand on. The theorems are neither dictator nor final guard. They are the ladder's honest report of its own reach, delivered from inside a medium that was never going to reach anyway, and that was the more important fact all along.

Appendix A: The sign convention and the foundational axioms

A.1 The sign convention

Read three unit rows, expressed in an orthonormal basis of the subspace they span, as pure quaternions a, b, c . Hamilton's product of two pure quaternions is $p \cdot q = -(p \cdot q) + p \times q$. Expanding the triple product and using that c is pure, $\text{Re}(a \cdot b \cdot c) = -(a \times b) \cdot c = -\det[a \cdot b \cdot c]$. The lock scalar used throughout this paper is $\lambda = \text{Re}(a \cdot b \cdot c)$, which is therefore *minus* the signed scalar triple product of the rows. Writing A for the matrix whose columns are a, b, c , the correlation matrix is $R = A^T A$ and $\det(R) = \det(A)^2 = \lambda^2$. The sign convention is stated here once and Section 5, Theorem 2, Table 1, and Appendix B all use it without variation. The identity $\det R = \lambda^2$ is insensitive to the choice, which is exactly why the choice must be stated rather than inferred.

A.2 Foundational axioms, softly veiled

The following axioms are drawn from a broader epistemic framework and are presented here as standalone principles, each independently motivated and independently testable within the native discipline.

Actuation as the mark of existence. To exist is to carry substrate-level energetic content and to charge a cost for any transition. This is theorem-grade for confined quantum existents by the kinetic-energy lower bound and the zero-point energy, both consequences of the canonical commutation relation, with mass-energy equivalence fixing positive rest energy and the rigorous third law barring a cooled-to-zero state.²⁷ Landauer's bound charges the cost of any irreversible erasure

and is measured rather than postulated.^{14,15} Its universal extension over all existents whatever rides a substrate-monist commitment and is premise-grade. It grounds the material self-presupposition argument of Section 6.5 and the physics leg of Section 6.4. It is not load-bearing on the type argument that establishes Box 1, nor on the formal results, which are pure linear algebra.

Grounding as the mark of formal being. To be formally determinate is to carry a determinate content that coincides with its own reflection across the ground-bearing involution. The fixed part is the decidable bridge; the reflected part is the residence the instrument reads. A purely self-dual proposition carries no residence and verifies nothing, the formal analogue of a tautology. Provability and computation are levels of access to a determinacy fixed at the ground rather than ingredients of it. This axiom is premise-grade and is the ground-first premise named in Section 8.

The premise-status of a foundation is itself a theorem. No consistent system on a base weaker than a posited foundation derives that foundation. The underivability is constitutive of foundation-hood and is a result about foundations rather than a foundation itself, so it seals without self-contradiction. This is why the ground-first premise is held premise-grade by design and not by concession, and it is the formal content of the claim that a foundation provable from its base would not be a foundation.

Appendix B: The battery, printed in full

Executing this file at seed 20260622 reproduces every numerical claim in this paper. It requires only numpy. No other input, dataset, or file is used.

PYTHON

```
import numpy as np
from itertools import product
from collections import Counter

def qmul(a, b):
    w1,x1,y1,z1 = a; w2,x2,y2,z2 = b
    return np.array([w1*w2-x1*x2-y1*y2-z1*z2,
w1*x2+x1*w2+y1*z2-z1*y2,
w1*y2-x1*z2+y1*w2+z1*x2,
w1*z2+x1*y2-y1*x2+z1*w2])

def kernel(M):
    M = np.asarray(M, float); N = M.shape[1]
    eps = 100.0*np.finfo(float).eps*N
    Mn = M - M.mean(1, keepdims=True); sd = Mn.std(1,
ddof=1, keepdims=True)
    if np.any(sd == 0): return 'open', None
    Mn = Mn/sd; Q = Mn/np.sqrt((Mn*Mn).sum(1,
keepdims=True))
    detR = float(np.linalg.det(Q @ Q.T))
    if detR <= eps: return 'broken', detR
    if np.linalg.cond(Q @ Q.T) >= 1e6: return 'open',
detR
```

```
return 'lock', detR

rg = np.random.default_rng(20260622); N = 24
t = np.linspace(0, 2*np.pi, N, endpoint=False)

# CHK.1 the two involutions
for nm, T in (('conjugation', np.diag([1.,-1,-1,-1])),
('antipodal', -np.eye(4))):
    w = np.linalg.eigvalsh(T)
    print(nm, 'residual', f'{np.linalg.norm(T@T-
np.eye(4)):.3e}',
'eigs', sorted(np.round(w,12).tolist()),
'dim Fix', int(np.sum(np.isclose(w,1))))

# CHK.2 substrate chirality
i,j,k = np.array([0.,1,0,0]), np.array([0.,0,1,0]),
np.array([0.,0,0,1])
print('Re(i*j*k)', f'{qmul(qmul(i,j),k)[0]:+.12f}')

# CHK.3/4 orientation-blindness on a QR-fixed basis
M4 = np.array([np.sin(t), np.cos(2*t), np.sin(3*t)]) +
0.03*rg.standard_normal((3,N))
Mn = M4 - M4.mean(1, keepdims=True); Mn = Mn/Mn.std(1,
ddof=1, keepdims=True)
Q = Mn/np.sqrt((Mn*Mn).sum(1, keepdims=True))
B = np.linalg.qr(Q.T)[0][:, :3].T
def lam(Qm):
    co = Qm @ B.T; q = [np.concatenate(([0.0], c)) for c
in co]
    return float(qmul(qmul(q[0], q[1]), q[2])[0])
Qr = np.diag([-1.,1.,1.]) @ Q
d0, dr = float(np.linalg.det(Q@Q.T)),
float(np.linalg.det(Qr@Qr.T))
l0, lr, ln = lam(Q), lam(Qr), lam(-Q)
print('det0', f'{d0:.12f}', '|d0-dr|', f'{abs(d0-dr):.
3e}')
print('lam0', f'{l0:.12f}', 'lamr', f'{lr:.12f}',
'ratio', f'{lr/l0:.6f}')
print('|lam^2-det|', f'{abs(l0*l0-d0):.3e}')
print('max|G(P)-G(~P)|', f'{np.max(np.abs(Q@Q.T - (-Q)@(-
Q).T)):.3e}',
'lam(~P)', f'{ln:.12f}')

# CHK.5 classification archetypes. Each negation is
STIPULATED contentless.
def roads(ph):
    L = np.array([np.sin(t+ph), np.cos(2*t+ph),
np.sin(3*t+ph)])
    A = np.array([[1.00,0.30,0.12],[0.20,1.00,0.26],
[0.15,0.22,1.00]])
    return A @ L + 0.05*rg.standard_normal((3,N))
flat = np.ones((3,N))
for nm, P, Np in (('THEOREM', roads(0.0), flat),
('GODEL', roads(0.7), flat),
```

```

        ('GHOST', roads(0.0), roads(1.3))) :
    vP, dP = kernel(P); vN, dN = kernel(Np)
    print(nm, vP, dP and f'{dP:.6f}', '| not-P', vN, dN
and f'{dN:.6f}')

# CHK.6 the denotation register
VARS=('p','q','r'); OPS=('&','|','>')
def ev(f,m):
    o=f[0]
    if o=='v': return m[f[1]]
    if o=='~': return not ev(f[1],m)
    if o=='&': return ev(f[1],m) and ev(f[2],m)
    if o=='|': return ev(f[1],m) or ev(f[2],m)
    return (not ev(f[1],m)) or ev(f[2],m)
def show(f):
    o=f[0]
    if o=='v': return f[1]
    if o=='~': return '~'+show(f[1])
    return '('+show(f[1])+o+show(f[2])+')'
def gen(d,rng):
    if d==0 or rng.random()<0.28:
        a=('v',VARS[rng.integers(0,3)])
        return ('~',a) if rng.random()<0.5 else a
    o=OPS[rng.integers(0,3)];
    f=(o,gen(d-1,rng),gen(d-1,rng))
    return ('~',f) if rng.random()<0.18 else f
def MI(xs,ys):
    n=len(xs);
    px,py,pxy=Counter(xs),Counter(ys),Counter(zip(xs,ys))
    return sum(((c/n)*np.log2((c/n)/((px[a]/n)*(py[b]/n))))
for (a,b),c in pxy.items())

BITS, WIDTH = 4, 64 # 9-symbol
alphabet -> 4 bits/symbol
rg6 = np.random.default_rng(20260622)
MODELS = [dict(zip(VARS,v)) for v in
product((False,True), repeat=3)]
forms = [gen(3, rg6) for _ in range(4000)]
forms = [f for f in forms if BITS*len(show(f)) <=
WIDTH] # admission filter
cf,cv,tr = [],[],[]
for f in forms:
    for m in MODELS:
        cf.append(WIDTH); cv.append(BITS*len(show(f)));
tr.append(ev(f,m))
print('formulas', len(forms), 'pairs', len(tr))
print('cost_fixed values', sorted(set(cf)))
print('I(cost_fixed;truth)', f'{MI(cf,tr):.12f}',
'H(truth)', f'{MI(tr,tr):.12f}')
print('true pairs', sum(tr), 'of', len(tr), '; closure
would force', len(tr)//2)
D = set(forms)
lackT = [f for f in forms if ('~', f) not in D]
A = sum(1 for f in lackT if f[0] == '~')

```

```

B = sum(1 for f in lackT if f[0] != '~' and
BITS*(len(show(f))+1) > WIDTH)
print('lacking negation', len(lackT), 'of', len(forms),
'; leading-neg', A, '; filter-cut', B, '; undrawn',
len(lackT)-A-B)
print('I(cost_var;truth)', f'{MI(cv,tr):.6f}',
f'= {100*MI(cv,tr)/MI(tr,tr):.4f}% of H(truth)')
L=np.array([len(show(f)) for f in forms],float)
T=np.array([sum(ev(f,m) for m in MODELS)/8 for f in
forms])
print('corr(length, truth-freq)', f'{np.corrcoef(L,T)
[0,1]:+.4f}')
print('cost_var(~P)==cost_var(P)+4 for all P:',
all(BITS*len(show('~',f))) == BITS*len(show(f))
+BITS for f in forms))

```

Extended theoretical connections

The fixed-locus-versus-diagonal distinction is not confined to arithmetic incompleteness, and three adjacent structures inherit it directly.

The unified diagonal. Cantor's uncountability, Tarski's undefinability, Russell's paradox, and Gödel's first theorem are one fixed-point phenomenon in a cartesian closed category.¹⁹ Read through the instrument, this is the statement that all four run on the reflection whose fixed locus has dimension zero, so all four measure a ladder and none reaches a ground. The categorical unification, far from supplying an escape, reinforces the placement: it shows the incompleteness engine is more structural and more robust than the arithmetic encoding suggested, and equally ground-blind in every instance.

The independence spectrum, partitioned by ground-presence. A true-but-unprovable arithmetic sentence and a set-theoretic independence such as the continuum hypothesis relative to a standard set theory are both independent of their systems, and the instrument sorts them into different classes. The arithmetic sentence reads grounded, a flat countable object that is rung-absent. The set-theoretic independence reads ungrounded, field-permitted in both directions on a supplied independence proof. The partition is by ground-presence rather than by cardinality, and it yields the concrete discipline named in Section 8: importing a maximality axiom to assign the continuum a determinate value treats a ghost as an imprint.

The division-algebra terminus. The three-reading instrument stops at three because the composition algebra stops at the quaternions. The next normed division algebra, the octonions, forfeits associativity: any three unit imaginaries not lying on a common Fano line have a nonzero associator, so no bracket-invariant verification survives there. The identity of a witnessing triple is a fact about the labelling and not about the algebra, and is therefore not stated here as a structural fact; the chart-free statement is the one above. The same boundary that fixes the instrument's axis count at three fixes the sphere on which the ground-bearing reflection lands, the unique associative group-sphere above dimension one. The terminus is a

closure rather than a limitation, and it is why no fourth independent reading is admissible.

Author's Provenance and Method Disclosure

This paper was developed under Trisduction, a verification and organizing discipline, not a source of results. Its conclusions rest solely on the standard results cited in the body. Trisduction is the method under which those results were decomposed, assembled, and audited. It adds them no warrant and claims no authorship over them.

Trisduction was used for fidelity. It forces a claim onto three independent axes so no single persuasive line carries it alone, requires every verdict to resolve to one of three states, sealed, broken, or open, each with a named failure mechanism, and attaches an explicit warrant grade, theorem, conditional, structural, or premise, to every claim so nothing is stated above its strength. Two errors it is built to catch are inflation, reading an internal lock as a proof, and circularity, reading a restatement of a claim as a derivation of it.

The root axiom, RA, is that to exist is to actuate: every physical existent carries a strictly positive energetic cost, grounded in the Heisenberg energy floor, the zero-point energy, and Landauer's bound. The formal root inside it is that to formally be is to be grounded, with provability and computation levels of access to a determinacy fixed at the ground rather than ingredients of it.

The key procedures, in brief. Orthogonal triaxial convergence: a proposition is split into three disjoint axes, formal-structural, empirical or dynamical, and registrational, verified by three separate instrument sets, agreement across the three the operational meaning of a seal. The Geometric Orthogonal Lock, GOL: the three axes are read as vectors and their independence is tested by the determinant of their correlation matrix, a determinant clear of zero a genuine three-dimensional lock, a collapse to zero one axis dissolving into the plane of the other two, a broken lock. The Convergence Dissolution Test, CDT: before the lock is read, any mass-bearing common cause the three axes might share is projected out, so an apparent convergence that traces to a shared source rather than to independent roads dissolves and carries no warrant, the lock computed on the residue that survives. The twelve-gate cascade: twelve directed failure screens, self-reference, frame-dependence, missing mechanism, and the rest, run in order, the first failure terminating the verdict with its mechanism named. The verdict issues in the three-state economy with its warrant grade attached, and where the argument stops short the open direction is named rather than filled.

Box G1 | The GOL kernel identity, proved

The lock is a determinant identity, not a metaphor. Let the three axes, after normalization, be unit vectors expressed in an orthonormal basis of the subspace they span and read as pure quaternions a, b, c , with norm one each. Hamilton's product of two pure quaternions is $p \cdot q = -(p \cdot q) + p \times q$, the real part the negative dot product and the imaginary part the cross product, checked on the units by $i \cdot j = k = i \times j$ and $i \cdot i = -1 = -(i \cdot i)$. The lock scalar is the real part of the triple product, $\lambda = \text{Re}(a \cdot b \cdot c)$.

Expanding, $a \cdot b = -(a \cdot b) + a \times b$, so $\text{Re}(a \cdot b \cdot c) = -(a \cdot b) \text{Re}(c) + \text{Re}(a \times b \cdot c)$. The first term is zero since c is pure, and the second is $-(a \times b) \cdot c$ by the same product rule, giving

$$\lambda = -(a \times b) \cdot c = -\det[a \ b \ c],$$

minus the signed volume of the parallelepiped the three axes span. Writing A for the matrix whose columns are a, b, c , the correlation matrix is $R = A^T A$, its entries the pairwise dot products, so

$$\det(R) = \det(A^T A) = \det(A)^2 = \lambda^2.$$

The kernel identity $\lambda^2 = \det(R)$ is that the squared signed volume equals the Gram determinant, with the quaternion triple product the machine that computes the volume. It is confirmed at machine precision, the residual $|\lambda^2 - \det(R)|$ at 1.110×10^{-16} on the recorded battery.

Four consequences fix the reading, and they are why the number can be trusted to say what the lock says. The determinant is bounded, $\det(R)$ in the interval from zero to one for unit axes, by Hadamard's inequality above and positive-semidefiniteness below, so the lock has a hard ceiling at one, the fully orthogonal frame, and a hard floor at zero. The floor is the break: $\det(R)$ equals zero exactly when the three axes are linearly dependent, one axis lying in the plane of the other two, which is the geometric content of a collapsed lock. The magnitude is orientation-blind: reflecting any axis sends A to a matrix of opposite determinant, flipping the sign of λ while $\det(R)$ equals λ^2 is unchanged, so the number certifies the dimensionality of the lock and never its truth-sign, which is read from the ordered axes and not from the scalar. And the magnitude is frame-invariant: rotating all three axes together is conjugation q to $u \cdot q \cdot \bar{u}$ on the imaginary quaternions, under which the real part is preserved and dot and cross products are covariant, so λ and $\det(R)$ depend on the configuration and not on the coordinate labels.

The axis count is three because the algebra forces it, not because three was chosen. The audit-composition law requires associativity, so iterated audits bracket the same way, and the absence of zero divisors, so nonzero warrants never compound to nothing. By Frobenius's theorem the only finite-dimensional associative division algebras over the reals are the reals, the complex numbers, and the quaternions, and three mutually orthogonal imaginary axes exist only in the quaternions, whose imaginary units i, j, k are the three axes. The next normed division algebra, the octonions with seven imaginary units, fails associativity on any three unit imaginaries not lying on a common Fano line, so it cannot carry the composition law and the

construction stops at three. The three axes are the imaginary part of the unique associative real division algebra that supports the audit.

This paper in particular. The three axes were instantiated only for the formal results: a fixed-locus dimension count, a determinant-against-triple-product identity, and a two-direction grounding test. The central theorem does not lock triaxially and is not claimed to. It is carried by a single analytic argument from the type of the derivation operator, doubled by one premise-disjoint argument that convicts grounding rather than closure, defended by a physics leg that establishes nothing, and carried outside the Gödelian window by a two-flank witness that applies it rather than corroborating it. The honest count of premise-disjoint supports is two, and only one is needed. The lock is the anchor battery of Section 7, executable in full from Appendix B at seed 20260622 over 24 contexts: an independent triad locks at det 0.999627146061 with the kernel identity closing at 1.110×10^{-16} , the reflected triad returns the identical determinant to bit-zero while λ inverts from +0.999813555650 to -0.999813555650, and the classification archetypes separate, a proven theorem at det 0.586297 and a true-but-unprovable sentence at det 0.588777 both reading grounded against a genuine independence result locking both directions at 0.617430 and 0.596375 and reading ungrounded. The denotation register reports an identity rather than a lock: at a fixed 64-bit frame the cost functional takes one value and its mutual information with truth is 0.000000000000 bits by constancy, while the variable-frame control returns 0.003816 bits against a truth entropy of 0.998664690729 bits, and that control is what forced the fixed-encoding scope onto the claim rather than leaving it unconditional. The load-bearing lemma is Presburger's completeness theorem, which does double duty: it establishes the jurisdiction containment and simultaneously bars the derivation containment that the containment invites. The verdict is a placement, sealed at structural grade with theorem-grade legs on the type argument, on the flanks, on the domain containment, on the fixed-locus invariance, and on the orientation-blindness identity, together with a premise-grade ground-first commitment that caps the boundary reading and not the arithmetic of the flanks. No new theorem is claimed; the paper reorganizes established results and applies them.

The canonical statement of the method, its axioms, and its executable batteries lives in the reference cited below, continuously updated at the same location; the version audited for this paper is the one pinned in the citation.

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Competing interests

The author declares no competing interests.

Data and code availability

Every numerical claim in this paper reproduces by executing the battery printed in full in Appendix B at seed 20260622 in double precision, requiring only numpy. No dataset, proprietary or otherwise, underlies any result, and no value is imported from another document. The classical results carrying the two flanks and the barred entailment are independently formalizable and are cited in full.

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